

Improving Step Accuracy in Specification-Driven Development

Classical engineering applied to LLM builds — decomposition, hard tests, and an optimized build graph

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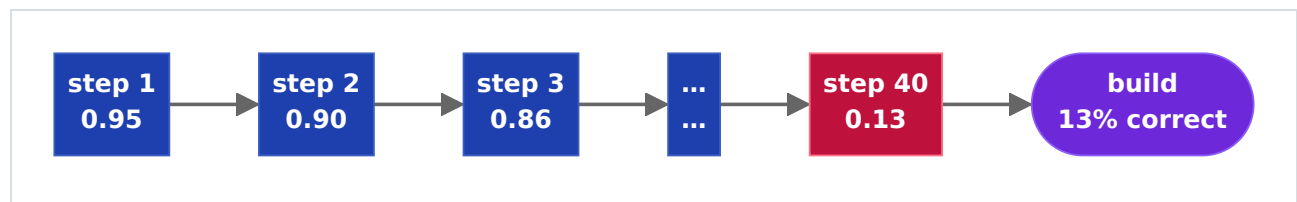
Abstract

Specification-driven development (SDD) builds software from a written specification by executing LLM build steps. Two failure modes dominate: error compounds across unverified steps, and model quality degrades as context grows. This paper applies classical engineering to both. Break the specification into stories by Agile decomposition. Put executable tests on each story. Relate the stories, with their prerequisite branding and enterprise stack rules, in a graph database and let test results gate the graph. Assemble each prompt from delimited specification blocks, and compress specifications to their contracts to optimize context. Every step is bounded, every step is verified, and the build repeats: the same specification produces the same working software. The method also supports change: edited specifications and change tickets enter the graph, and the correct build context is calculated dynamically.

Keywords: specification-driven development, LLM code generation, test-driven development, Agile decomposition, graph database, prompt stacking, context compression

1. The Accuracy Problem

A build is a chain of dependent steps. If each step is correct with probability p , the chain is correct with probability p^n . At $p = 0.95$, forty steps deliver a correct build less than 13% of the time.



Unverified error compounds: each step multiplies the survival probability of the build.

The arithmetic applies to every multi-step form:

- An agent chaining n tool calls.

- A pipeline of n build prompts.
- One giant prompt. The model still generates the application serially — file after file, decision after decision. The steps are internal and invisible, but they are steps, and each one can be wrong. Nothing about a single prompt removes the exponent; it only removes the ability to verify between steps.

The second failure mode is context growth. A 250,000-token specification fits inside a modern context window, but fitting is not comprehension. As context grows, measured model behavior degrades [4]:

- Constraints stated early are missed.
- Similar sections are conflated.
- Material is weighted by position, not relevance.

A step prompted with the full specification starts with a lower p than the same step prompted with only what it needs — before any compounding begins.

Both failure modes yield to classical engineering:

- Break the specification into stories with hard tests, and surface missing information as questions instead of guesses.
- Determine prerequisite branding and enterprise stack rules.
- Optimize story grouping for context and quality.
- Build reproducible software.

2. Simplification #1: Agile Epic Decomposition

The standard way to break an Epic into stories is Agile. The LLM understands Agile because it is well documented. The specification is the Epic. Epic → features → stories. Decompose until each story builds in one bounded step.

Every story specification ends with four sections:

Section	Contents
Programmatic Acceptance	Executable assertions that define done (§3)
User Acceptance	Checks only a human can honestly judge
Guardrails	What the software must never do
Open Questions	Missing information, returned to the product owner

A story that cannot be given Programmatic Acceptance is not yet a story; decompose further.

Open Questions is the feedback channel. Decomposition exposes what the specification does not say; those gaps are written as questions and answered by a human before the story builds. The mechanism for collecting answers varies and is not specified here. Ambiguity is resolved by a person, never guessed by the model.

3. Simplification #2: Test-Driven Development for Story Quality

Test-driven development supplies the per-story discipline: write the test before the code; the test suite — not the author — decides when the work is done.

- Programmatic Acceptance is written at decomposition time, as Pythonic, executable assertions: concrete checks against files, routes, return values, and observable behavior.
- Prose criteria ("the import should work correctly") are not acceptance. An assertion that cannot execute cannot gate a build.
- The assertions enter the build prompt as the step's success condition. The task changes from interpreting an instruction to satisfying declared assertions.
- After the step, the test suite runs outside the model. The model is never asked whether it finished; the suite passes or the story fails.



Each story builds, then its test suite executes. Failure loops back; a verified story unlocks the next.

4. Relating Features and Stories in a Graph Database

Stories are not a list; they are a graph. Features and stories are nodes; dependencies are edges; the graph is stored as plain text alongside the specification.

The graph also carries the build's prerequisites. Branding files (palette, typography, document standards) and enterprise stack rules (language, framework, and platform conventions) are determined up front and related to the stories that need them. These are the rules a reproducible build requires: without them, two builds of the same specification diverge on every convention the specification does not state.

The graph does four jobs:

1. **Ordering.** The runnable frontier — stories whose dependencies have all passed — is computable by inspection. Build order is a property of the data.
2. **Gating.** A story runs only when its dependencies have verified. A failed story blocks its dependents; the defect is caught where it was created and repaired locally.
3. **Containment.** With a test suite at every edge, no unverified chain exceeds length one. The §1 arithmetic collapses from p^n to p per step.
4. **Enabling build optimization.** The graph carries the token estimates and shared-context relationships that make the optimizations of §7 computable.

5. How to Assemble a Stack of Prompts

A build prompt is assembled, not written. Each step stacks the exact files the graph relates to it — story specifications, branding, and stack rules — each wrapped in a unique delimiter naming the file and its role. The delimiters make each block unambiguous to the LLM.

```
<header explaining layout>

<unique_delimiter filename="FEATURE-Import.md" role="implements">
  ...specification content...
</unique_delimiter>

<unique_delimiter filename="python.md" role="stack">
  ...stack rules...
</unique_delimiter>

<prompt instructions>
```

6. Compression: Optimizing Context

Many stories do not need the full version of their prerequisites. For example, a user interface screen needs only a summary of the web routes it calls — not the schemas, migrations, and design rationale behind them.

This is the Builder/User distinction. The builder of a feature needs its full specification. Users of the feature need only the contract: routes, class names, method signatures, typed parameters, one-line summaries. Each specification file gains a compact derivative containing only that contract; the builder stacks the full file, every user stacks the derivative.

Compression attacks both §1 failure modes: context per step falls, and what remains is exactly what the step consumes.

7. Optimization: Repeatable Quality Builds

The pieces compose:

Piece	Contribution
Decomposition (§2)	Every step is small enough to be accurate
Test suites (§3)	Every step proves itself before anything depends on it
Graph (§4)	Order is computed; errors are contained at edges
Stack assembly (§5)	Every prompt is a deterministic function of declared files
Compression (§6)	Every prompt carries contracts, not bulk

The graph makes this optimizable algorithmically. Example: 30 stories, each prompt 20,000 tokens, half of it shared stack rules and architecture. Built one at a time, the shared material is injected 30 times: 600,000 tokens total. Group the stories three per step and the shared material is injected 10 times: 400,000 tokens — a third of the

build cost removed by regrouping alone. The grouping is computed from the graph's token estimates and shared-context edges, tuned for both cost and build quality.

The method supports change. An edited specification or a change ticket enters the graph, invalidates only the stories that depend on it, and the correct build context for the rework is calculated dynamically. The rebuild is the affected subgraph, not the application.

Repeatability is the sum. The specification, the graph, the test suites, and the assembly rules determine every prompt and every acceptance decision. The same inputs produce the same verified software.

Drydock [1] is the reference implementation of this method.

8. Conclusion

This is classical engineering applied to a new build tool:

- Break the specification into stories small enough to be accurate.
- Put an executable test suite on every story.
- Determine prerequisite branding and enterprise stack rules.
- Relate stories, features, and rules in a graph and let the test suites gate it.
- Optimize story grouping for context and quality; compress what is merely consumed.
- Surface what is missing as questions for a human.
- Build reproducible software.

Error stops compounding, context stops confusing, and the build repeats.

References

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